

Problem 121. Let P be one of the intersection points of two circles with centres O_1, O_2 . A common tangent touches the circles at A, B respectively. Let the perpendicular from A to the line BP meet O_1O_2 at C .

Prove that AP is perpendicular to PC .

Problem 122. In an acute-angled triangle ABC with $\angle CAB > 60^\circ$, let

- H be its orthocentre;
- M, N be two points on AB, AC respectively, such that $\angle HMB = \angle HNC = 60^\circ$;
- O be the circumcentre of $\triangle HMN$;
- D be a point on the same side with A of BC such that $\triangle DBC$ is an equilateral triangle.

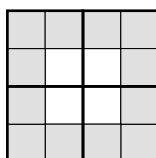
Prove that H, O, D are collinear.

Problem 123. Find all polynomials $P(x)$ with integer coefficients such that, for all integers a and b , $P(a+b) - P(b)$ is a multiple of $P(a)$.

Problem 124. Let $n > 1$ be an even positive integer. A $2n \times 2n$ grid of unit squares is given, and it is partitioned into n^2 contiguous 2×2 blocks of unit squares. A subset S of the unit squares satisfies the following properties:

- (1) for any pair of squares A, B in S , there is a sequence of squares in S that starts with A , ends with B , and has any two consecutive elements sharing a side; and
- (2) in each of the 2×2 blocks of squares, at least one of the four squares is in S .

An example for $n = 2$ is shown below, with the squares of S shaded and the four 2×2 blocks of squares outlined in bold.



In terms of n , what is the minimum possible number of elements in S ?